

AegisVault: Institutional ZK RWA Collateral & Selective Compliance

Official Technical Architecture & Product Specification | Midnight Network

Target Network:	Midnight Preprod Testnet (0x01)	Language / Compiler:	Compact v0.19 (BLS12-381)
Live Website:	aegisvalutmoonlight.netlify.app	GitHub Repo:	github.com/ayush-tech3/AegisVault
Demo Video:	youtu.be/GK1J3Dq58_8 (1080p)	Contract Address:	0x4e8a...1928 (Preprod)

1. Executive Summary & Selected Moonshot Track

AegisVault solves the fundamental barrier preventing institutional capital from entering DeFi: **the total lack of balance sheet and portfolio privacy on public transparent blockchains**. By leveraging Midnight Network's zero-knowledge dual-state architecture and the Compact smart contract language (v0.19), AegisVault allows institutional borrowers to deposit off-chain tokenized Real-World Assets (US Treasury Bills, AAA Corporate Notes, Commercial Real Estate Equity) into shielded vaults and generate confidential loans with mathematically proven 150% over-collateralization invariants.

Selected Idea from Official Midnight List:

- **Private Allowlist Access:** Borrowers prove membership in a verified accredited investor KYC/AML Merkle tree without disclosing their identity, wallet address, or specific whitelist leaf index.
- **Confidential Credentials:** Proves that the borrower's locked collateral meets or exceeds the required 150% over-collateralization ratio without revealing the exact dollar balance, asset allocation, or transaction history.

2. Problem Statement vs. AegisVault ZK Solution

Traditional Public Blockchains (Ethereum, Solana)	AegisVault on Midnight Network
■ Balance Sheet Exposure: Every wallet address, collateral balance, and liquidation threshold is visible to competitors and front-runners.	■ 100% Shielded Balance: Only 32-byte cryptographic commitments are stored on-chain. Zero leakage of actual portfolio size.
■ Regulatory Catch-22: Forced to choose between total public transparency or total opacity (which violates AML/KYC securities laws).	■ Rational Privacy: Programmable selective disclosure allows borrowers to grant cryptographic viewing keys to regulators (SEC/FINRA/ESMA).
■ Double-Borrowing Risk: Inability to prevent double-pledging of collateral without publishing account identities.	■ Deterministic ZK Nullifiers: Single-use nullifiers mathematically prevent double-borrowing attacks with zero identity disclosure.

3. Dual-State Architecture (Public Ledger vs. Private Witness)

AegisVault enforces a strict mathematical boundary between data visible to public observers and data retained exclusively inside the borrower's local Midnight Lace wallet:

Public Ledger State (Visible On-Chain)	Private Witness (Kept in Local Wallet)
• collateralCommitments: Map<Bytes[32], Commitment> • spentNullifiers: Set<Bytes[32]> (Anti-Double-Borrow) • activeLoans: Map<String, LoanRecord> (Debt, Status) • auditorDisclosures: Map<String, AuditRecord> • totalProtocolBorrowed: Uint<64>	• borrowerSecret: Bytes[32] (Private signing key) • collateralValueUSD: Uint<64> (Actual deposit amount) • assetType: Uint<8> (Treasury Bills, Bonds, Real Estate) • salt: Bytes[32] (Cryptographic hiding entropy) • merkleProof: Vector<Bytes[32], 16> (KYC Inclusion)

4. Compact Smart Contract Circuits (`contract/src/index.compact`)

Circuit Name	Core Invariant & Purpose
<code>depositShieldedCollateral</code>	Computes commitment hash from private witness and records 32-byte hash in public registry.
<code>borrowShielded</code>	Validates over-collateralization ($\geq 150\%$), verifies KYC Merkle inclusion, emits single-use nullifier, and opens loan.
<code>repayLoan</code>	Marks loan as Repaid, reduces protocol debt tally, and unlocks collateral without leaking borrower identity.
<code>grantAuditorDisclosure</code>	Generates an encrypted viewing key bound to the designated regulator's public key (SEC/FINRA/ESMA).

5. Automated Test Suite (8/8 Passing Tests)

AegisVault includes an automated test suite verifying all circuits, mathematical constraints, and security invariants:

- **Test 1:** Successfully registers Shielded RWA Collateral Commitment.
- **Test 2:** Rejects duplicate collateral commitments.
- **Test 3:** Generates valid ZK proof and borrows when collateral ratio $\geq 150\%$.
- **Test 4:** Rejects loan proof generation if collateral ratio is below 150% (undercollateralized).
- **Test 5:** Rejects borrower who is not part of the accredited investor Merkle whitelist.
- **Test 6:** Prevents double-borrowing using deterministic ZK nullifiers.
- **Test 7:** Successfully repays an active loan and reduces protocol debt.
- **Test 8:** Grants selective auditor disclosure viewing key without leaking data to public ledger.